
ROYAL FITNESS CLUB

Fitness Foundations Certificate

MODULE 8

Final Assessment & Certificate

Duration	20 minutes
Course Code	CS_B1
Level	Beginner
Format	Study Guide PDF

Part of the Royal Fitness Club Professional Certification Program.

ASSESSMENT INSTRUCTIONS

How to Complete This Assessment

- This is a 25-question multiple-choice assessment covering all 7 modules
- Each question has one correct answer (A, B, C, or D)
- Passing score: 70% (18/25 correct)
- You may refer to your module notes — this is an open-book assessment
- Complete in order; do not skip and return (track time: target 20 minutes)
- Answer key is provided at the end of this document

SECTION A: Anatomy & Physiology (Questions 1–7)

SECTION B: Training & Programme Design (Questions 8–14)

SECTION C: Nutrition (Questions 15–20)

SECTION D: Recovery, Mindset & Common Mistakes (Questions 21–25)

1. Which type of bone is found at the end of long bones and provides a spongy lattice that distributes compressive load?

- A) Compact (cortical) bone
- B) Cortical endplate
- C) Cancellous (trabecular) bone
- D) Periosteum

2. The Sliding Filament Theory of muscle contraction states that:

- A) Myosin and actin filaments both shorten during contraction
- B) Actin filaments slide over myosin, shortening the sarcomere
- C) ATP is not required for cross-bridge detachment
- D) Type I fibres generate more force than Type IIx fibres

3. Which of the four rotator cuff muscles is responsible for shoulder internal rotation?

- A) Infraspinatus
- B) Supraspinatus
- C) Teres minor
- D) Subscapularis

4. The Henneman Size Principle states that:

- A) Larger motor units are always recruited first
- B) Motor units are recruited from smallest to largest as force demands increase
- C) Type IIx fibres are recruited before Type I at low intensities
- D) Motor unit synchronisation decreases with training

5. Which energy system is the primary fuel source during a 200m sprint?

- A) Oxidative phosphorylation
- B) Creatine phosphate (ATP-PCr) system
- C) Glycolytic (anaerobic) system
- D) Beta-oxidation of fats

6. VO2 max is best improved by training in which intensity zone?

- A) 50–60% HRmax (Zone 1 — recovery)
- B) 65–75% HRmax (Zone 2 — aerobic base)
- C) 85–95% HRmax (Zone 4 — VO2 max zone)
- D) 100%+ HRmax (Zone 5 — anaerobic)

7. Growth hormone (GH) is primarily released:

- A) Immediately before exercise
- B) During light aerobic exercise
- C) During N3 deep sleep (first cycle of the night)
- D) After eating a high-carbohydrate meal

8. Which rep range is best supported by research for maximising muscle hypertrophy?

- A) 1–5 reps at 90–100% 1RM
- B) 6–20 reps at 65–85% 1RM
- C) 20–50 reps at 30–50% 1RM
- D) No single range; intensity must always exceed 85% 1RM

9. A beginner performs 3 sets of 8 squats at 60kg and completes 10 reps on the final set. Two sessions in a row this happens. According to the 2-for-2 rule, they should:

- A) Continue at 60kg until they can consistently complete 12 reps
- B) Immediately increase to 70kg to challenge themselves more
- C) Increase to 62.5kg at the next session
- D) Reduce to 55kg to practise perfect technique at lower load

10. The SAID principle stands for:

- A) Strength Adaptation In Determined individuals
- B) Specific Adaptation to Imposed Demand
- C) Standard Assessment for Incremental Development
- D) Sequential Adaptation In Development

11. Which training split provides the highest muscle group training frequency per week for a beginner training 3 days per week?

- A) Body part split (chest day, back day, leg day)
- B) Push/Pull/Legs split
- C) Full-body split
- D) Upper/Lower split

12. The RAMP warm-up protocol stands for:

- A) Resistance, Activation, Mobility, Power
- B) Raise, Activate, Mobilise, Potentiate
- C) Range, Aerobic, Muscle, Preparation
- D) Readiness, Activation, Movement, Performance

13. During a deload week, the primary goal is to:

- A) Achieve maximum strength by testing 1RM lifts
- B) Rest completely and do no physical activity
- C) Reduce training volume to allow fatigue dissipation and enable supercompensation
- D) Switch to purely cardiovascular training to maintain calorie burn

14. Which type of exercise produces the greatest Excess Post-Exercise Oxygen Consumption (EPOC)?

- A) 60-minute steady-state walking
- B) 30-minute moderate-intensity cycling (60% VO₂ max)
- C) Heavy compound resistance training and HIIT intervals
- D) Light stretching and yoga flow

15. The primary driver of muscle protein synthesis (MPS) is activation of which intracellular complex?

- A) AMPK
- B) mTORC1
- C) FOXO3
- D) NF- κ B

16. A 30-year-old male, 80kg, 178cm trains 4 days/week. His BMR (Mifflin-St Jeor) is approximately 1850 kcal. What is his TDEE?

- A) 1850 kcal
- B) 2220 kcal
- C) 2868 kcal
- D) 3330 kcal

17. Which fatty acid type is associated with strong anti-inflammatory effects and should be prioritised in an athlete's diet?

- A) Saturated fatty acids
- B) Trans fatty acids
- C) Omega-6 polyunsaturated fatty acids
- D) Omega-3 polyunsaturated fatty acids (EPA + DHA)

18. The optimal protein intake per meal to maximally stimulate muscle protein synthesis is approximately:

- A) 5–10g of any protein source
- B) 20–40g of a complete, high-leucine protein source
- C) 50g+ regardless of leucine content
- D) 10–15g immediately before training only

19. Dietary fat intake should NOT fall below approximately what percentage of total calories to protect hormonal health?

- A) 5%
- B) 10%
- C) 15%
- D) 25%

20. A urine colour described as "dark yellow / apple juice" indicates:

- A) Optimal hydration
- B) Mild dehydration requiring 250–500ml additional fluid
- C) Severe dehydration requiring medical attention
- D) Excess vitamin supplementation with no action needed

21. DOMS (Delayed-Onset Muscle Soreness) is primarily caused by:

- A) Lactic acid accumulation remaining in the muscle 24–48 hours after exercise
- B) Microscopic eccentric muscle fibre damage and subsequent inflammatory response
- C) Dehydration and electrolyte imbalance in muscle tissue
- D) Insufficient warm-up causing muscle spasm

22. Which supplement has the strongest evidence base and is most universally recommended for strength and power athletes?

- A) BCAAs (branched-chain amino acids)
- B) Fat burners/thermogenics
- C) Creatine monohydrate (3–5g/day)
- D) Testosterone boosters

23. The testosterone:cortisol ratio is most likely to become unfavourable when:

- A) Training sessions are kept under 45 minutes
- B) Training is performed in a fully rested, well-fed state
- C) High-intensity training sessions consistently exceed 90 minutes

D) Resistance training is combined with 20 minutes of aerobic work

24. Cold water immersion (ice bath) post-workout is BEST suited to which scenario?

A) A beginner in their first 12-week hypertrophy block

B) An athlete competing in back-to-back games requiring rapid recovery

C) A powerlifter training a new 1RM PR

D) Anyone trying to maximise muscle gain in an off-season

25. According to the SMART goal framework, which of the following is the best example of a specific, measurable goal?

A) "Get fit and feel healthier this year"

B) "Exercise more consistently starting next month"

C) "Increase my 5km run time from 32 minutes to 26 minutes within 12 weeks"

D) "Lose some weight and build muscle by summer"

ANSWER KEY

Question	Answer	Key Concept
Q1	C	Cancellous bone
Q2	B	Sliding filament — actin slides
Q3	D	Subscapularis — IR
Q4	B	Henneman size principle
Q5	C	Glycolytic system — 200m
Q6	C	Zone 4 85–95% HRmax
Q7	C	N3 deep sleep — GH pulse
Q8	B	6–20 reps hypertrophy
Q9	C	2-for-2 rule → +2.5kg
Q10	B	SAID principle
Q11	C	Full body = highest freq for 3-day
Q12	B	RAMP warm-up
Q13	C	Deload = reduce volume/fatigue
Q14	C	HIIT+compounds = max EPOC
Q15	B	mTORC1 drives MPS
Q16	C	$1850 \times 1.55 = 2868$ kcal
Q17	D	Omega-3 EPA+DHA anti-inflammatory
Q18	B	20–40g leucine-rich protein
Q19	C	15% min fat for hormones
Q20	B	Dark yellow = mild dehydration
Q21	B	DOMS = eccentric damage + inflammation
Q22	C	Creatine monohydrate
Q23	C	>90 min = cortisol rises, T:C unfavourable
Q24	B	Back-to-back competition
Q25	C	SMART — specific 5km in 12 weeks

ROYAL FITNESS CLUB
FITNESS FOUNDATIONS CERTIFICATE

This is to certify that

has successfully completed the

Fitness Foundations Certificate — CS_B1

demonstrating mastery of anatomy, exercise science, programme design, nutrition, recovery, mindset, and applied fitness fundamentals.

Instructor Signature

Date